

8. Which of the following statements is not true:

- (a) Presence of water vapour is highly variable in the lower atmosphere.
- (b) The zone of maximum temperature is located along the equator
- (c) Frigid zones are located in both the hemispheres between the polar circles and the poles.
- (d) Jet streams are high altitude winds affecting the surface weather conditions.

U.P.P.C.S. (Pre) 2002

Ans. (b)

The average, temperature decreases from the Equator towards higher latitudes. It may be noted that the highest temperature is never recorded at the Equator, instead, it is recorded at 20°N. So, it can be said that the zone of maximum temperature is not located at the equator. Therefore, statement (b) is not correct, other statements are correct.

9. The highest temperature on the earth is recorded at -

- (a) Equator
- (b) 10° North Latitude
- (c) 20° North Latitude
- (d) 25° North Latitude

M.P.P.C.S. (Pre) 2010

Ans. (c)

See the explanation of the above question.

10. Clear sky nights are cooler than cloudy sky nights due to

- (a) condensation
- (b) radiation
- (c) induction
- (d) conduction

39th B.P.S.C. (Pre) 1994

Ans. (b)

The cloudy night contains more water vapor than a clear night. The heat emitted from the Earth's surface is trapped by the clouds and emitted back towards the earth. As a result a cloudy night seems warmer than clear a night. If the sky is clear, heat emitted from the Earth's surface freely escapes into space, resulting in colder temperatures.

11. With reference to the Earth's atmosphere, which one of the following statements is correct?

- (a) The total amount of insolation received at the equator is roughly about 10 times of that received at the poles.
- (b) Infrared rays constitute roughly two-thirds of insolation.

- (c) Infrared waves are largely absorbed by water vapour that is concentrated in the lower atmosphere.
- (d) Infrared waves are a part of visible spectrum of electromagnetic waves of solar radiation.

I.A.S. (Pre) 2023

Ans. (c)

The largest amount of energy that reaches the Earth's surface is in short wavelengths. The energy that the Earth receives is referred to as insolation, or incoming solar radiation. Notably, shortwave solar energy can pass through the atmosphere quite easily. Before reaching the surface of the Earth, the incoming solar energy travels through the atmosphere. A large portion of the near-infrared light is absorbed by water vapor, ozone, and other gases in the troposphere. Due to this reason assertion (c) is true.

The Cyclones

*A cyclone is a large scale air mass that rotates around a strong center of low atmospheric pressure. Cyclones rotate anticlockwise in the northern hemisphere and clockwise in the southern hemisphere.

*Depending on the location cyclones can be categorized into two types – tropical cyclones and temperate cyclones.

(1) Tropical Cyclones – (Between tropics of Cancer and Capricorn). Tropical cyclones are known by different names in different places. In northern and Northwestern part of Australia they are called **Willy-Willy**, in China, they are called **typhoons**, in the USA they are known as **hurricanes** and in India they are called **cyclones**. In the Great Plains of USA, strong cyclones are formed as a result of Cold polar winds coming in the contact with hot air brought by Gulf streams. These cyclones are called **tornadoes**. **Tornado Alley** is the name given to the region of the Great Plains affected by Tornadoes. The places located in the Tornado Alley are – **Mississippi-Missouri Valley, Texas, Oklahoma, Kansas and Nebraska**. Tornadoes are common in the entire tropical and subtropical region but are very prominent in the Caribbean Sea and North America Region. *Fujita scale is used to measure the intensity of tornadoes.

*Hurricane is a tropical cyclone affecting the regions of Texas, Mexico, Florida and West Indies. **Saffir Simpson scale** is used to measure potential property damage caused by a hurricane.

*Typhoon is also a tropical cyclone that originates in the South China Sea. **Nargis** was a tropical cyclone responsible for heavy destruction in Myanmar.

(2) Temperate Cyclones – These cyclones originate in the mid latitudes. They are often round, building or wedge-shaped.

These cyclones are found between **35°-65°** latitudes in both hemispheres. Under the influence of the westerlies blow from west to east.

***Anticyclones** have high pressure at their center, around which air circulates. *The pressure decreases from centre towards the outer periphery due to which air blows from the center outwards.

*Anticyclones are clockwise in the northern hemisphere and anticlockwise in the southern hemisphere.

1. Consider the following statements :

1. Jet streams occur in the Northern Hemisphere only.
2. Only some cyclones develop an eye.
3. The temperature inside the eye of a cyclone is nearly 10°C lesser than that of the surroundings.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 2 only (d) 1 and 3 only

I.A.S. (Pre) 2020

Ans. (c)

Jet streams are currents of air high above the surface of Earth. They move generally west to east at altitudes of about 8 to 15 kilometers. Their speed usually varies from 129 to 225 kilometers per hour, but they can reach 443 km/hr. They occur in both hemispheres - northern and southern.

The "eye of a cyclone is an area of comparatively light winds and fair weather found at the center of a severe tropical cyclone. The temperature inside the eye of a cyclone is higher than at the surrounding area (the eye temperature may be 10°C warmer at an altitude of 12 km than the surrounding environment, but only 0-2°C higher at the surface).

2. Assertion (A) : The surface winds spiral inwards upon the center of the cyclone.

Reason (R) : Air descends at the center of the cyclone.

Code :

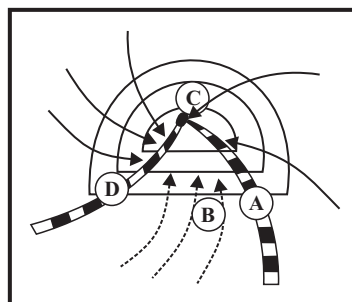
- (a) Both (A) and (R) are true, and (R) is the correct explanation of (A)
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

I.A.S. (Pre) 2002

Ans. (a)

A cyclone is a large scale air mass that rotates around a strong center of low atmospheric pressure. They are usually characterized by inward spiraling winds around the center of the cyclone. A cyclone center is the area of lowest atmospheric pressure. Air descends at the center of the cyclone. Thus, both (A) and (R) are true and (R) is the correct explanation of (A).

3. Which one of the areas marked as A, B, C and D in the given figure of the cyclone, witnesses heavy torrential short duration rainfall accompanied by thunderstorms?



- (a) A (b) B
(c) C (d) D

I.A.S. (Pre) 1999

Ans. (d)

The different parts of the cyclone have different weather phenomena due to the differences in air mass and temperature. The air tends to rise and causes a drop in pressure. When two different air masses meet, the boundary zone between them is called a front. The warm air glides over the cold air and a sequence of clouds appears in the sky ahead of the warm front (A) and causes precipitation. The cold front (D) approaches the warm air from behind and pushes the warm air up. As a result, cumulus clouds develop along the cold front. In cold front area, the rising temperature produces strong upward winds. These winds carry water droplets upwards, where they freeze and fall down again. The swift movement of the falling water droplets along with the rising air creates lightning and sound. This event is a thunderstorm.

4. The Tornadoes are very strong tropical cyclones originating in:

- (a) Caribbean Sea (b) China Sea
(c) Arabian Sea (d) Black Sea

U.P.P.C.S. (Pre) 1996

Ans. (a)

Tornadoes are tropical cyclones. They are small in size but are immensely powerful and destructive. A tornado is a dark funnel-shaped cloud that reaches from the sky to the ground. Normally tornadoes occur in all tropical and subtropical regions, but the Caribbean Sea and the American region are most prone to them.

5. Which of the following would you generally associate with Tornadoes ?

- (a) Gulf of Mexico (b) China Sea
(c) Indian Ocean (d) U.S.A.

U.P. Lower (Spl) (Pre) 2009

Ans. (d)

See the explanation of the above question.

6. Which one of the following regions of the U.S.A. is known as "Tornado Alley"?

- (a) Atlantic Seaboard (b) Pacific Coast
(c) Mississippi Plains (d) Alaska

U.P.P.C.S. (Pre) 2005

Ans. (c)

In the great plains of USA, when a polar cold air mass meets with a warm humid tropical air mass a frightful cyclone develops, known as a 'Tornado'. Tornado Alley is an area in plains of the mid-USA that has a very high number of destructive tornadoes. Tornado Alley mainly includes the states of Texas, Oklahoma, Kansas and Nebraska as well as the Mississippi and Missouri Valley.

7. To measure the intensity of Tornadoes we use-

- (a) Mercalli scale (b) Fujita Scale
(c) Saffir -simpson scale (d) Richter scale

U.P. Lower Sub. (Pre) 2009

Ans. (b)

The Fujita-Pearson Scale more popularly known as the F-scale is used to measure the intensity of a tornado. The scale was named after Ted Fujita.

8. Which of the following area is more susceptible to typhoon cyclone?

- (a) Australia (b) China-sea
(c) Asia (d) America

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

A typhoon is a mature tropical cyclone that develops in the mid-northern hemisphere of the China Sea between 100°-180°E longitude.

9. Match List-I with List-II and select the correct answer using the codes given below the lists :

List-I

- A. Australia
B. China
C. India
D. U.S.A.

List-II

1. Hurricanes
2. Willy-Willy
3. Typhoons
4. Cyclones

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	3	4	1
(c)	3	2	1	4
(d)	4	3	2	1

U.P.P.C.S. (Mains) 2005

Ans. (b)

A tropical cyclone is known by different names in different parts of the world. It is called a 'Hurricane' in the USA, 'Typhoon' in China. In North and North-West Australia it is called Willy-Willy and in India it is called Cyclones.

10. Willy-Willy is:

- (a) A type of tree grown in temperate regions
(b) A wind that blows in a desert
(c) A tropical cyclone of the north-west Australia
(d) A kind of common fish found near Lakshadweep islands

U.P.U.D.A./L.D.A. (Pre) 2001

I.A.S. (Pre) 1995

Ans. (c)

See the explanation of the above question.

11. Match List – I with List – II and select the correct answer from the codes given below :

List – I

- A. Willy-Willy
B. Hurricanes
C. Typhoons
D. Baguio

List – II

1. U.S.A.
2. Australia
3. Philippines
4. China

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	2	1	4	3
(c)	1	2	4	3
(d)	2	1	3	4

U.P.P.C.S. (Pre) 2014

Ans. (b)

The correct match is as follows :

Willy-Willy	-	Australia
Hurricane	-	U.S.A
Typhoons	-	China
Baguio	-	Philippines

12. Match List-I with List-II and select the correct answer from the codes given below:

List-I (Tropical Cyclones)	List-II (Country)
A. Baguios	1. Australia
B. Hurricanes	2. China
C. Typhoons	3. Philippines
D. Willy-Willies	4. United States of America

Code :

A	B	C	D
(a) 3	4	1	2
(b) 3	4	2	1
(c) 2	3	4	1
(d) 2	1	3	4

U.P. P.C.S. (Pre) 2020

Ans. (b)

See the explanation of the above question.

13. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I (Different name of Tropical cyclone)	List-II (Country)
A. Willy-Willies	1. Philippines
B. Taifu	2. Australia
C. Baguio	3. Japan
D. Hurricanes	4. U.S.A.

Code :

A	B	C	D
(a) 3	4	1	2
(b) 2	3	4	1
(c) 1	3	2	4
(d) 2	3	1	4

U.P.P.C.S. (Pre) 2019

Ans. (d)

List-I (Different name of Tropical cyclone)	List-II (Country)
Willy-Willies	Australia
Taifu	Japan
Baguio	Philippines
Hurricanes	U.S.A.

As per the options, the correct answer would be options (d).

14. Which of the following scales is used to measure the damage by a hurricane?

- (a) Saffir-Simpson Scale (b) Mercalli Scale
(c) Fujita Scale (d) Richter scale

U.P.P.C.S. (Mains) 2016

Ans. (a)

The Saffir Simpson Scale measures the wind speed of the hurricane. Wind speed is measured on a scale of 1-5 to give an estimate of potential property damage and flooding along the coast.

15. A sudden fall of Mercury in Barometer indicates;

- (a) Fair weather (b) Stormy weather
(c) Cold weather (d) Hot weather

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

A sudden fall of the Barometer indicates westerly winds followed by a violent storm. Thus, option (b) is the correct answer.

16. Which of the following cyclones did not affect India?

- (a) Aila (b) Nargis
(c) Thane (d) Neelam

U.P. U.D.A./L.D.A. (Pre) 2013

Ans. (b)

The tropical cyclones *Aila*, *Thane*, and *Neelam* affected India but the *Nargis* cyclone did not affect India. *Nargis* cyclone mainly affected the region of Myanmar.

17. In the South Atlantic and South-Eastern Pacific regions in tropical latitudes, cyclone does not originate. What is the reason?

- (a) Sea surface temperatures are low
(b) Inter-Tropical Convergence Zone seldom occurs
(c) Coriolis force is too weak
(d) Absence of land in those regions

I.A.S. (Pre) 2015

Ans. (a)

The most proximate reason for the absence of cyclones in the South Atlantic and South Eastern Pacific region in tropical latitudes is the low temperature of the sea surface.

18. As a result of global warming the frequency and severity of which of the following are increasing due to global warming?

- (a) Cyclones only (b) Storms only
(c) Hurricanes only (d) All of the above

U.P.P.C.S (Pre) 2018

Ans. (d)